

NL-to-SQL Lakehouse

DuckDB star schema + semantic layer + text-to-SQL

March 11, 2024

Headline Result

50M-row lakehouse; 100% NL->SQL execution accuracy

Prepared for: Sai Veda

Publishing account: Nikeshk834

Project Context

A DuckDB lakehouse over a partitioned-Parquet star schema, with a semantic layer and a deterministic, offline text-to-SQL engine. Ask a business question in English - "top 5 product categories by revenue", "which region has the lowest revenue?", "monthly revenue in 2023" - and the system links schema entities, compiles validated SQL through the semantic layer, and executes it out-of-core against 50M fact rows.

This PDF dossier is intentionally visual and interview-ready: it captures the business problem, the technical architecture, the validation evidence, and the committed repository artifacts in a format that reads like a delivered project brief.

Repository slug

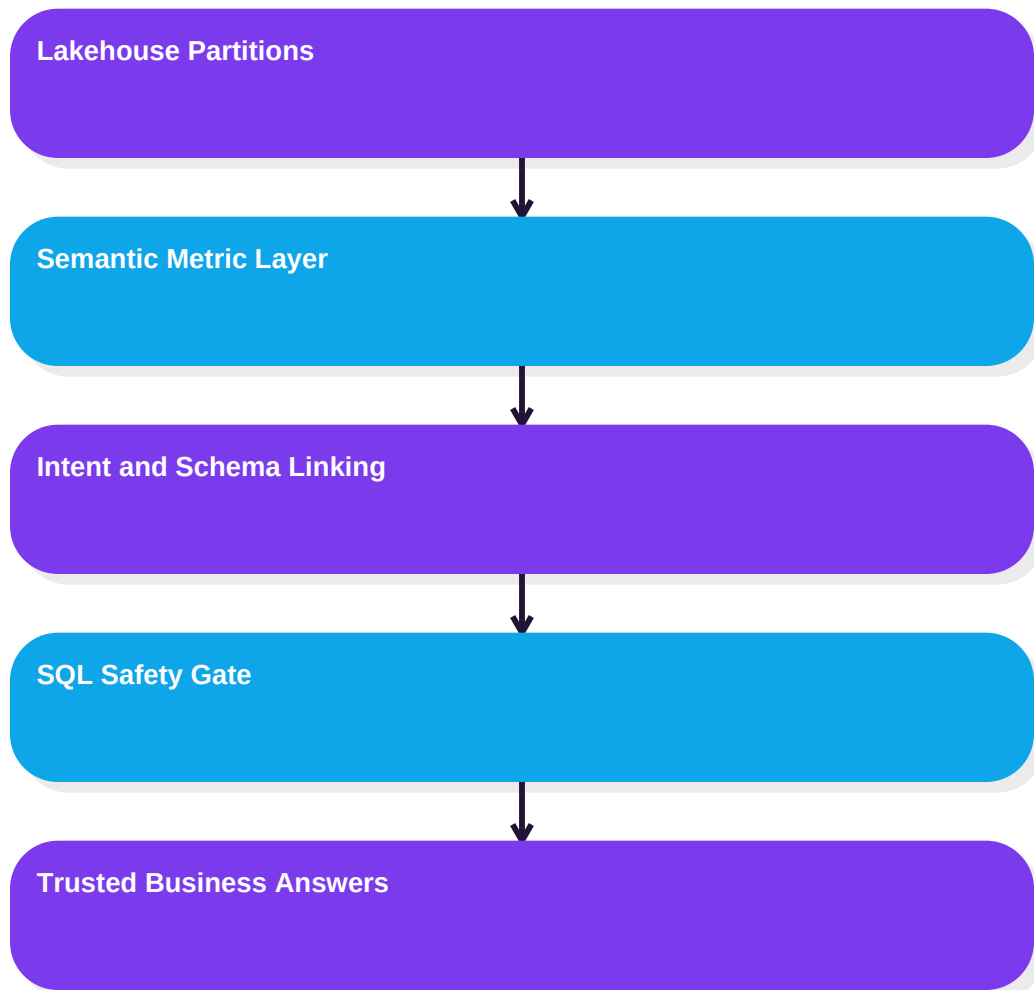
06-nl-to-sql-lakehouse

Validation index

36 tests

Architecture Storyboard

The semantic layer and SQL guardrails are the load-bearing parts that keep natural-language analytics trustworthy.



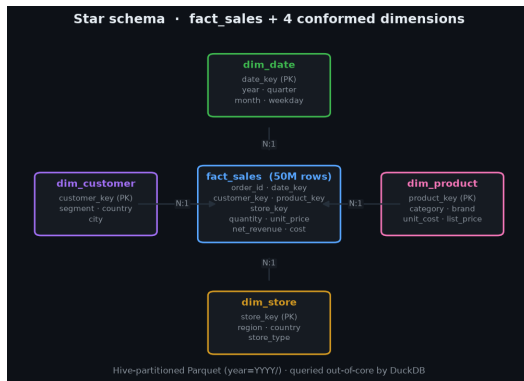
High-level focus area: The semantic layer and SQL guardrails are the load-bearing parts that keep natural-language analytics trustworthy.

Validation And Visual Evidence

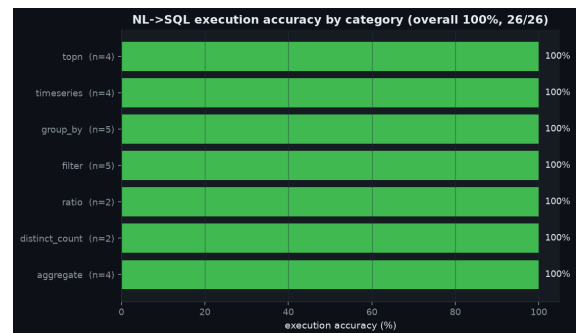
50M-row lakehouse; 100% NL->SQL execution accuracy

36 passing tests

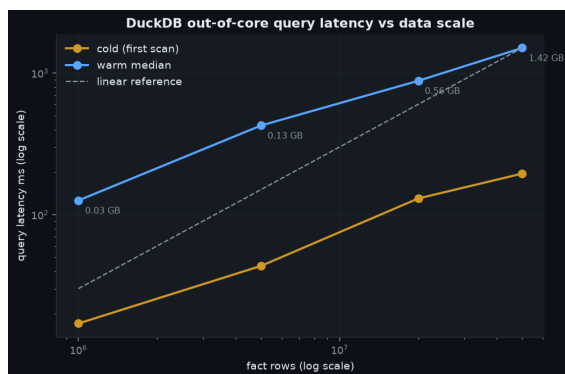
March 11, 2024



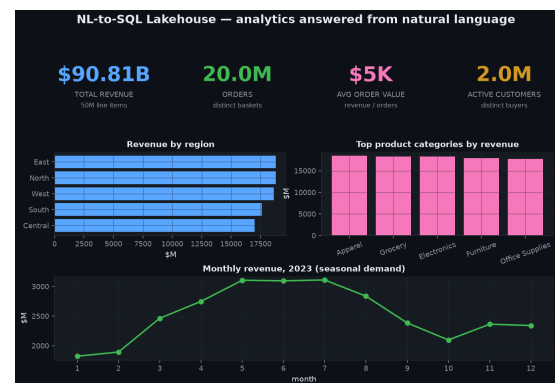
Star schema (data model)



NL->SQL execution accuracy by category



Query latency vs data scale (log-log)



Analytics answered from natural language

Repository Handoff

Prepared for: Sal Veda

Publishing account: Nikeshk834

Project date: March 11, 2024

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Committed deliverables

- README.md - product overview, quickstart, and measured headline numbers
- ARCHITECTURE.md - detailed design and scale trade-offs
- tests/ - automated validation suite
- benchmarks/ - measured run artifacts and result tables
- assets/ - generated dashboards, charts, and screenshots
- PROJECT_DOCUMENT.pdf - styled project brief for sharing and review

This document is intentionally more visual than the repository Markdown. It is meant to read like a polished project brief: business framing first, architecture second, proof third, and delivery scope last.